The University of Faisalabad

Computer Organization and Assembly language

Course: Computer Organization and Assembly Language (Lab) - CS-318

Array Practice in EMU8086

Objective:

Learn to handle arrays and loop through them.

Theory:

- Arrays are declared in .DATA segment.
- Use a loop to process array elements.

Example Program: Sum of Array

```
ORG 100h


DATA SEGMENT

arr DB 10, 20, 30, 40, 50

n DB 5

DATA ENDS


CODE SEGMENT

ASSUME CS:CODE, DS:DATA
```

```
START:
    MOV AX, DATA
    MOV DS, AX

    MOV CX, n       ; Number of elements
    MOV SI, 0       ; Array index
    MOV AL, 0       ; Sum accumulator

SUM_LOOP:
    ADD AL, arr[SI]
    INC SI
    LOOP SUM_LOOP

    ; AL now contains sum
    MOV AH, 4Ch
    INT 21h
CODE ENDS
END START
```

Program Logic

1. Set up the data segment.
2. Initialize CX as loop counter, SI as array index, AL as sum accumulator.
3. Loop through all elements of the array:

Add each element to AL

Move to next element

4. After loop, AL holds the sum of array elements.

5. Exit program using DOS interrupt.

Lab Exercise

- Reverse an array and display its elements.
